

RESEARCH ARTICLE

MTA-LSTMNet: A Multi-Task Attention LSTM Architecture for Continuous Spatio-Temporal Drought Forecasting in Andhra Pradesh

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ABSTRACT Meteorological and hydrological droughts are known to have complex spatio-temporal dynamics that limit the effectiveness of traditional early warning systems. Machine learning models used for drought forecasting are mainly focused on solving binary classification tasks rather than reflecting the severity of drought and the temporal nature of climatic disruptions. Addressing these limitations, this study introduces the novel Multi-Task Attention LSTM (MTA-LSTMNet) architecture that enables concurrent predictions of SPEI-3, SPEI-6, and SPEI-12 drought severities. Training is based on a 26-year long (1995-2020) high-resolution spatial data set covering the state of Andhra Pradesh, India, with a historical window of six months and a collection of twelve Essential Climate Variables (ECVs). The presented architecture is characterized by a unique approach to integrating the effects of macro-oceanic teleconnections (Mean Sea Levels) and micro-atmospheric aerosol disruptions ($PM_{2.5}$, CO_2). Specifically, a new Temporal Attention mechanism has been developed to calculate the inertia of the regional water cycle, showing that the model's accuracy is 63.7% dependent on the immediate preceding conditions, while still having a meaningful memory tail up to six months ago. The presented model showed highly competitive results for all three timelines – R^2 of 0.919, 0.916, and 0.914, respectively, which is a significant improvement over traditional methods such as XGBoost and Random Forest. In terms of the feature importance, permutation analysis showed that the Arabian Sea Level was the global variable impacting regional droughts the most, while also identifying $PM_{2.5}$ concentration as a local factor. Finally, the spatio-temporal mapping proved the model capable of accurately reflecting asymmetric recovery in the coastal areas and detecting rain shadow drought vulnerability zones such as Rayalaseema.

INDEX TERMS Multi-task learning, long short-term memory (LSTM), spatio-temporal forecasting, standardized precipitation evapotranspiration index (SPEI), temporal attention mechanism, explainable AI (XAI), Andhra Pradesh.

I. INTRODUCTION

Meteorological and Hydrological Droughts refer to creeping natural phenomena defined by persistent shortages in rainfall and high atmospheric evaporation needs [28]. Due to climate change and increasing anthropogenic greenhouse emissions, climate anomalies occur more often and with more intensity; hence, posing catastrophic threats to global agriculture and economics as a whole [37]. In geographically

diverse locations such as Andhra Pradesh, India, where tropical coasts gradually transform into dry arid zones like Rayalaseema, the need for highly localized early warning systems remains imperative [8].

Drought index calculation involves using statistical indicators such as the Standardized Precipitation Index (SPI) and Standardized Precipitation Evapotranspiration Index (SPEI). Although SPI uses only precipitation data, SPEI uses PET as well, making it a better indicator of drought occurrence because it accounts for temperature differences and global warming issues better [31]. While the two are

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quite mathematically solid indices, the historical approach for forecasting involved the use of auto-regression models that failed to model multidimensional nonlinear relationships between climatic factors [11].

The past decade witnessed a significant revolution as ML and DL models have replaced statistics-based methods because they offer much better modeling of climatic spatiotemporal dependencies [29]. Among all types of neural architectures, Recurrent Neural Networks (RNNs) with LSTM structures showed a promising ability to interpret time series data; however, there are several profound structural weaknesses revealed by a careful literature review:

A. THE BINARY CLASSIFICATION PROBLEM

In dealing with extremely imbalanced climatological data, most state-of-the-art models currently predict drought occurrence by reducing the problem to a simple binary classification (Drought/Non-Drought) [2]. For example, a recently published paper applied a Single Objective Scorer (SOS) algorithm along with the Gradient Boosting Classifier in predicting drought occurrence without estimating any index value [2]. Although such methodology allows for precise balance between precision and recall, it ultimately destroys the continuous scale of drought severity itself. Climate anomalies do not occur through binary on/off switching, but by way of gradual physical changes.

B. ONE-TIMELINE LIMITATION

Most models confine their predictions to one specific timeline (for instance, SPEI-3). Such an approach neglects the mathematical enforcement of water cycle mechanics, wherein short-term agricultural drought dictates the momentum of groundwater abundance [38].

C. STATIC EXPLAINABILITY (XAI)

Recent approaches have attempted to implement XAI methods such as SHAP (SHapley Additive exPlanations) in order to understand the drought's dependency on the relevant variables [4]. Such an approach is entirely unaware of time. A static SHAP score is incapable of explaining the time-delayed effect of oceanic anomalies months before drought formation.

D. IGNORANCE OF MICRO-ATMOSPHERIC DISRUPTORS

Current predictive modeling is mainly based on conventional meteorological data (temperature, precipitation), and the presence of macro-oceanic anomalies [32]. Yet, recent research demonstrates that micro-atmospheric disruptors, namely $PM_{2.5}$ aerosols and regional CO_2 spikes, considerably disturb the local radiation balance and evaporative transpiration processes [33].

In light of the aforementioned problems, this study proposes a novel architecture based on the MTA-LSTMNet algorithm. Leveraging the power of a vast, multimodal data set spanning 26 years in Andhra Pradesh, this methodological

approach treats drought prediction as a multidimensional regression problem instead of a binary classification.

The key contributions of this research include:

- **Multi-Task Continuous Regression:** A single unified architecture is proposed to predict exact continuous drought severity scores in agriculture (SPEI-3), hydrology (SPEI-6), and groundwater levels (SPEI-12).
- **Dynamic Temporal Explainability:** A novel Temporal Attention Mechanism is integrated into the LSTM architecture, allowing us to measure when the model weights its predictions in a 6-month rolling time window, accurately capturing the “recency bias” and “memory tail” properties of the climate system.
- **Spatial Aerosol-Ocean Level Integration:** An extremely high-dimensional 3D spatial tensor is developed using 12 Essential Climate Variables (ECVs), providing a mathematical proof for the dominance of the Arabian Sea level and capturing the local interference of $PM_{2.5}$.
- **Spatio-Temporal Visualizations:** Unlike traditional tabular metrics, our model is applied to map the geographical movement of drought severity scores across the grid of Andhra Pradesh, demonstrating its effectiveness as a localized early warning system.

II. SYSTEMATIC LITERATURE REVIEW

To conduct an unbiased evaluation of drought forecasting models, we followed a systematic literature review approach according to PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) methodology [1].

A. PRISMA SEARCH STRATEGY AND SELECTION CRITERIA

An extensive search was performed in numerous scientific databases such as IEEE Xplore, ScienceDirect, Web of Science, and Scopus. Boolean parameters specific to deep learning-based drought forecasting and spatiotemporal attention mechanisms were used in the search process. A total of 842 papers were initially identified. After removing duplicate references, 656 titles and abstracts were screened. Papers focusing exclusively on non-meteorological forecasting or using archaic autoregressive models were rejected during the eligibility phase.

For a rigorous evaluation, full texts were critically assessed based on three stringent criteria: (1) the model's ability to forecast continuous severity indices, (2) multi-scale predictive capabilities, and (3) incorporation of temporal XAI. Only 38 core papers met the inclusion criteria. The flow diagram of PRISMA processes is presented in Fig. 1.

B. LITERATURE SYNTHESIS AND IDENTIFIED RESEARCH GAPS

Synthesizing the literature provides valuable insights into the evolutionary pattern of existing methodologies, but it also unveils the consistent presence of a “binary classification paradigm” in Table 1.

Contemporary models' reliance on classifiers instead of regressors renders them incapable of detecting severity

TABLE 1. Summary of existing literature and identified research gaps.

Reference	Objective	Algorithms Used	Limitations	Proposed Solution in MTA-LSTMNet
Sundararajan et al. [2]	Meteorological drought prediction in Tamil Nadu.	Gradient Boosting, SOS Algorithm	Binary classification strictly; fails to output continuous severity.	Continuous regression outputs for exact severity decimals.
Dikshit et al. [3]	Short-term spatio-temporal drought forecasting.	Random Forest (RF)	Single-timescale isolation; lacks deep temporal memory mechanisms.	Parallel multi-task heads (SPEI-3, 6, 12) to capture inertia.
Lundberg et al. [4]	General model interpretability.	SHAP (Explainable AI)	Static explainability; mathematically blind to sequential time delays.	Dynamic Temporal Attention mapping specific historical months.
Fung et al. [11]	Review of ML in drought forecasting.	ARIMA, SVM, Standard ANN	Autoregressive models fail to map non-linear climate dynamics.	Deep stacked LSTM sequence encoders.
Ali et al. [6]	Incorporate climate and seasonality into forecasting.	Extreme Learning Machine (ELM)	Static machine learning; lacks sequential memory for rolling windows.	Utilizes sequence-to-sequence LSTMs to preserve temporal momentum.
Kratzert et al. [29]	Rainfall-runoff hydrological modeling.	Standard LSTM	Single-task focus; does not capture parallel agricultural/groundwater dynamics.	Multi-Task Learning (MTL) to jointly optimize the full water cycle.
Shi et al. [30]	Spatio-temporal precipitation nowcasting.	Convolutional LSTM	Focuses on immediate short-term rain events; misses slow-onset creeping drought.	Explicitly models continuous multi-month severity gradients (up to 12 months).
Chandra [40]	Spatial and temporal drought forecasting.	Standard RNNs / LSTMs	Suffers from informational bottlenecks; forgets deep historical anomalies.	Integrates an Attention Mechanism to preserve the 6-month memory tail.
Zhang et al. [7]	Analyzing meteorological sensitivity to $PM_{2.5}$.	Statistical ML analysis	Evaluates aerosols as target variables rather than predictive disruptors.	Integrates $PM_{2.5}$ directly into the 12-feature 3D input tensor.
Bhuiyan et al. [8]	Spatiotemporal drought analysis in Andhra Pradesh.	Statistical correlations	Retrospective analysis only; lacks a forward-looking predictive engine.	Deploys a deep learning forward-forecasting spatial architecture.
Reddy & Symeonakis [10]	Assessment of drought in the Rayalaseema region.	Standard SPI / SPEI metrics	Relies on manual index calculation without predicting future spatial vulnerabilities.	Provides continuous, automated spatial progression mapping of the region.
Sen et al. [39]	Deep learning applications in environmental sciences.	Various Deep Neural Networks	Broad conceptual overview; lacks a localized framework for regional forecasting.	Highly dimensioned, region-specific continuous regression methodology.

gradients necessary for efficient water management [3]. In addition, their exclusive use of static XAI techniques (e.g., SHAP) results in “temporal blindness” among researchers [2].

To predict the temporal gap between oceanic anomalies and subsequent droughts, dynamic sequential-to-sequential attention mechanisms should be employed [5]. Such methods are notably absent in regional Indian drought forecasting research. This paper specifically aims to address the aforementioned isolated gaps.

III. STUDY AREA AND DATA ENGINEERING

A. STUDY AREA: TOPOGRAPHICAL AND CLIMATIC DYNAMICS

The main geographical region chosen as the object of research is the southeastern state of Andhra Pradesh, located between $12^{\circ}41'$ and $19^{\circ}07'$ N latitude and $76^{\circ}50'$ and $84^{\circ}40'$ E longitude [8]. As such, Andhra Pradesh can be characterized as a very complicated climatological lab. On one hand, the eastern borders of the state are marked by a long coastline along the Bay of Bengal, which makes the state prone to cyclone activity [9]. On the other hand, the southwestern interior – specifically Rayalaseema region – is located within the rain shadow of the Western Ghats mountain range [10].

This geographic duality leads to an incredible degree of variation within the area when it comes to regional moisture availability. Hence, to make the most of the information, predictive algorithms need to account for historical climatic momentum at the spatial grid point level.

B. MULTI-SOURCE DATASET CONSTRUCTION AND GATHERING

For accurate representation of the multiple factors leading to drought, a continuous spatial dataset was created based on 26 years of observations (January 1995 to December 2020). The dataset was gathered at $0.5^{\circ} \times 0.5^{\circ}$ spatial resolution. In contrast to the conventional models based solely on rainfall and temperature, the present study constructs a high-dimensional feature space consisting of 12 Essential Climate Variables. Datasets from various sources with different native spatial resolutions have been aligned using bilinear interpolation to the common $0.5^{\circ} \times 0.5^{\circ}$ master grid.

The data gathering protocol involved querying the NASA POWER API [12] for meteorological fluxes, while macro-oceanic teleconnections were scraped from the IMF Climate Dashboard [13]. Aerosol optical depth data ($PM_{2.5}$) was acquired via the ACAG spatial grids [14], and greenhouse gas arrays from the EDGAR v6.0 database [15]. Table 2 details the variables.

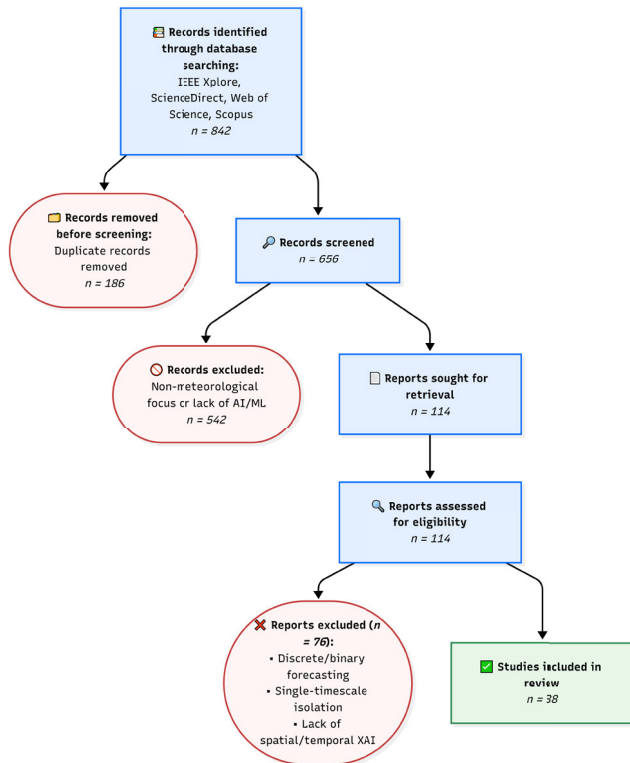


FIGURE 1. PRISMA flow diagram detailing the systematic literature search and selection process for DL-based drought forecasting.

TABLE 2. Multi-source dataset and essential climate variables (1995–2020).

S.No.	Climatic Variable	Data Repository
1	Precipitation (PRECTOT)	NASA POWER [12]
2	Mean Temperature (T2M)	NASA POWER
3	Dew Point Temp (T2MDEW)	NASA POWER
4	Wind Speed at 10m	NASA POWER
5	Shortwave Radiation	NASA POWER
6	Arabian Sea MSL	IMF Dashboard [13]
7	Bay of Bengal MSL	IMF Dashboard
8	Indian Ocean MSL	IMF Dashboard
9	Particulate Matter ($PM_{2.5}$)	Wash. Univ. ACAG [14]
10	Carbon Dioxide (CO_2)	EDGAR v6.0 [15]
11	Cyclical Month (Sine)	Mathematical Encoding
12	Cyclical Month (Cosine)	Mathematical Encoding
13	SPEI-3, SPEI-6, SPEI-12	CSIC Global SPEI [16]

C. DATA IMPUTATION AND FEATURE SCALING

Due to satellite sensor calibration errors over the 26-year period, approximately 1.4% of the grid data contained missing values. These gaps were resolved utilizing a spatio-temporal linear interpolation algorithm [25].

Deep Neural Networks exhibit severe mathematical sensitivity to unscaled feature spaces. To stabilize gradient descent, a Min-Max scaling transformation was applied independently to all input features, projecting the values into a uniform [0, 1] tensor space:

$$X_{scaled} = \frac{X - X_{min}}{X_{max} - X_{min}} \quad (1)$$

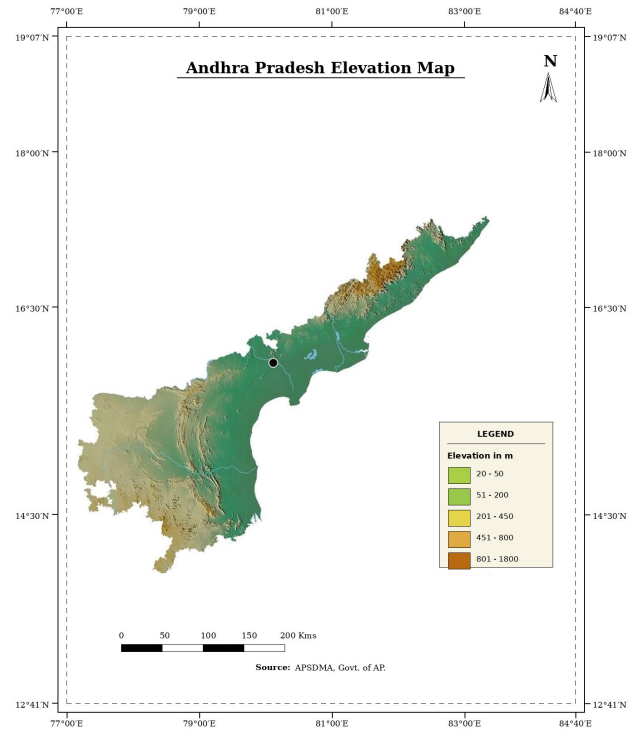


FIGURE 2. Elevation map of Andhra Pradesh (EMAP). The topographical profile highlights the severe contrast between the state's vulnerable eastern coastline and the arid, rain-shadow isolation of the southwestern Rayalaseema region.

In Eq. (1) – the symbol X is used for representing the original raw measurement of a certain climate variable (CO_2 concentration, Wind Speed, etc.). Parameters X_{min} and X_{max} stand for minimum and maximum possible value of the variable X , respectively, recorded during the course of the training dataset (26 years). The scaling of input variables in accordance with Eq. (1) prevents large-scale features with huge numerical ranges from overpowering the neural network. Continuous SPEI target variables were deliberately kept outside the scope of this scaling transformation, so that the model output remained completely interpretable.

D. CYCLICAL TIME ENCODING AND 3D TENSOR WINDOWING

As it may result in the loss of seasonal continuity, the categorical month variable needed to be geometrically transformed into its two-dimensional continuous representation by projecting it onto a circle:

$$Month_{sin} = \sin\left(\frac{2\pi \times M}{12}\right) \quad (2)$$

$$Month_{cos} = \cos\left(\frac{2\pi \times M}{12}\right) \quad (3)$$

In Equations (2) and (3) M stands for integer month number ($M \in [1, 12]$), whereas 2π represents 360° rotation

in radians. Thus, by dividing 2π by 12 we get the angular separation of 1/12-th of a full rotation. The trigonometric functions then translate it into 2D coordinates, thus guaranteeing that the Euclidean distance between subsequent months – e.g., Month 12 and Month 1 – remains the same and thus preserving the continuity of the seasonality momentum.

Finally, the scaled data was subjected to a sliding-window algorithm with a temporal window of $t = 6$ months. This transformed the flat 2D tabular data into a robust three-dimensional input tensor of shape (195762, 6, 12), representing over 195,000 unique six-month rolling historical windows isolated by geographic coordinates.

IV. PROPOSED METHODOLOGY: MULTI-TASK ATTENTION LSTM

To accurately model the continuous spatio-temporal dynamics of meteorological drought, this study proposes a novel MTA-LSTMNet network. Unlike traditional tree-based ensembles that process isolated, flat tabular vectors [2], the proposed architecture is specifically engineered to ingest 3D historical tensors, dynamically weight the temporal sequence, and simultaneously output three distinct drought severity indices.

As explicitly detailed in Fig. 3, the network topology is sequentially divided into four distinct phases: (1) Spatio-Temporal Input Formulation, (2) Shared LSTM Feature Extraction, (3) Temporal Attention Weighting, and (4) Parallel Multi-Task Regression Decoding.

A. SPATIO-TEMPORAL INPUT FORMULATION

The foundational input to the architecture (Phase 1) is the 3D spatio-temporal tensor engineered during the data preprocessing phase. Rather than evaluating instantaneous climate conditions, the network models physical momentum. For any given prediction target at month T , the network ingests a 6-month rolling historical window ($t - 6$ to $t - 1$) encompassing all 12 Essential Climate Variables (ECVs), including macro-oceanic sea levels and micro-atmospheric aerosols. This formulation results in a strictly dimensioned input tensor $X \in \mathbb{R}^{N \times 6 \times 12}$ (where N represents the total number of localized geographic samples), which is continuously fed into the sequential encoder.

B. SHARED LSTM SEQUENCE ENCODER

The foundational layer (Phase 2 in Fig. 3) is designed to extract deep temporal momentum from the input tensor X . Standard Recurrent Neural Networks (RNNs) suffer from vanishing gradients when processing extended climatological sequences [19]. To mitigate this, a stacked LSTM architecture is employed [20].

In terms of visual decomposition shown in Fig. 4, the inner design of each LSTM cell is designed to carefully control the process of long-term climatic memory transfer. During any temporal instant t , the cell receives three different inputs: the present high-dimensional meteorological tensor x_t , the past

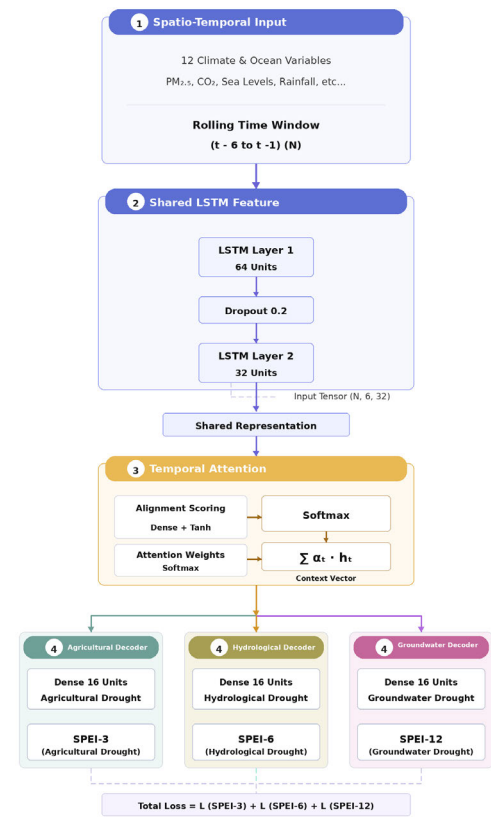


FIGURE 3. The proposed MTA-LSTMNet architecture, illustrating the flow from 3D spatio-temporal tensors through the shared sequence encoder, the novel temporal attention mechanism, and the parallel multi-task continuous regression decoders.

short-term hidden state h_{t-1} , and the past long-term memory cell state C_{t-1} .

The concatenated inputs $[h_{t-1}, x_t]$ first traverse the **Forget Gate**, where a sigmoid activation (σ) applies a pointwise numerical mask between 0 and 1. This operation mathematically determines which outdated historical anomalies (e.g., dissipated cyclonic events) should be erased from the continuous cell state. Simultaneously, the **Input Gate** (sigmoid) and the **Candidate State** ($\tanh$) compute the relevance and magnitude of the new incoming climatic data. The pointwise multiplication ($\times$) of these two streams creates a filtered update vector, which is then added ($+$) to the forgotten state to generate the newly stabilized long-term memory, C_t .

Finally, the **Output Gate** determines which specific portion of this newly updated cell state is actively relevant for the immediate drought severity prediction. The new cell state C_t is pushed through a $\tanh$ activation to scale the values between -1 and 1 , and subsequently multiplied by the output gate's sigmoid mask to yield the final hidden state h_t . This output is then propagated vertically to the subsequent stacked LSTM layer, and horizontally to the next chronological month in the 6-month sequence.

At each temporal step $t \in [1, 6]$, the LSTM cell updates its hidden state h_t and cell state C_t using a system of

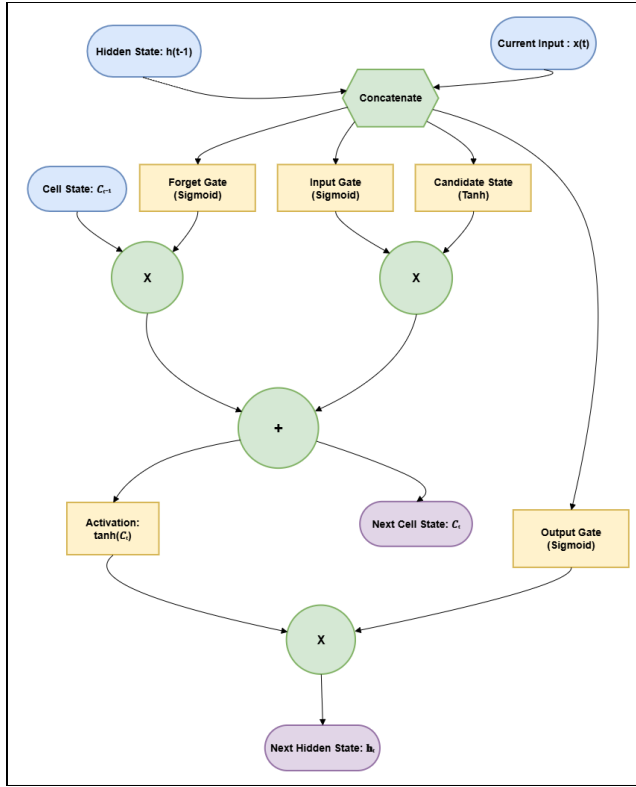


FIGURE 4. Internal architecture of the utilized LSTM cells, detailing the mathematical routing of the forget, input, and output gates alongside the hidden (h_t) and cell (C_t) states.

regulating gates. The forget gate f_t dictates which historical climate information is discarded:

$$f_t = \sigma(W_f \cdot [h_{t-1}, x_t] + b_f) \quad (4)$$

The input gate i_t and candidate cell state $\tilde{C}_t$ determine what new climatic anomalies are stored:

$$i_t = \sigma(W_i \cdot [h_{t-1}, x_t] + b_i) \quad (5)$$

$$\tilde{C}_t = \tanh(W_C \cdot [h_{t-1}, x_t] + b_C) \quad (6)$$

The cell state is then updated, and the output gate o_t filters the hidden state representation:

$$C_t = f_t * C_{t-1} + i_t * \tilde{C}_t \quad (7)$$

$$h_t = o_t * \tanh(C_t) \quad (8)$$

In Equations (4) through (8), W_x and b_x designate the trainable weight matrices and bias vectors for the respective gates, while σ and $\tanh$ represent the standard non-linear activation functions.

To ensure robust feature extraction and prevent overfitting, the network utilizes two shared LSTM layers. The first LSTM layer consists of 64 units. Crucially, it is configured to return the full temporal sequence ('Return Sequences = True'), mathematically expanding the tensor into a high-dimensional feature space of $\mathbb{R}^{N \times 6 \times 64}$. A Dropout layer (rate = 0.2) is applied to penalize the co-adaptation of hidden units [21].

Layer two LSTM (32 nodes) handles this sequence, once again providing the full sequence, resulting in the final "Shared Representation" tensor of $\mathbb{R}^{N \times 6 \times 32}$. It is essential to preserve the temporal dimension ($T = 6$) for the subsequent attention operation.

C. TEMPORAL ATTENTION MECHANISM (XAI)

In standard single-layer LSTM models, only the last hidden state h_T is fed to the output layer, introducing an information bottleneck [22] that obliterates the "memory trail" of the hydrological cycle. For this reason, an innovative Temporal Attention Mechanism (Phase 3) was designed. This layer works as an adaptive XAI module, analyzing the SR and generating a predictive relevance score α_t for each historical month.

First, the attention weights e_t are computed by using a dense layer with hyperbolic tangent as the activation function:

$$e_t = \tanh(W_a h_t + b_a) \quad (9)$$

The Softmax function is applied solely along the temporal axis to normalize the scores to form an attention probability distribution (summing up to 1.0):

$$\alpha_t = \frac{\exp(e_t)}{\sum_{k=1}^T \exp(e_k)} \quad (10)$$

In turn, the Context Vector c , that is, the weighted climate history, can be calculated as the dot product between the attention coefficients and the LSTM hidden states:

$$c = \sum_{t=1}^T \alpha_t h_t \quad (11)$$

Here, W_a and b_a are the attention layer parameters. In this way, the model can indicate precisely *when* a certain drought cause had the highest impact on predicting the current severity level.

D. PARALLEL MULTI-TASK REGRESSION DECODERS

The integrated Context Vector c can be seen as the network's holistic representation of the regional water cycle. Instead of forcing the network to provide one particular timescale, the network uses Multi-Task Learning (MTL) [23] by dividing c into three separate decoders (Task Heads 1, 2, and 3).

Every parallel decoder has a 16-unit Dense layer with ReLU activation and acts as a unique feature detector of its drought time series, followed by a linear output neuron for continuous decimal prediction:

$$\hat{y}_{SPEI-3} = W_{out3}(\text{ReLU}(W_{d3}c + b_{d3})) + b_{out3} \quad (12)$$

$$\hat{y}_{SPEI-6} = W_{out6}(\text{ReLU}(W_{d6}c + b_{d6})) + b_{out6} \quad (13)$$

$$\hat{y}_{SPEI-12} = W_{out12}(\text{ReLU}(W_{d12}c + b_{d12})) + b_{out12} \quad (14)$$

The strong parameter sharing through the encoder results in an extremely significant regularization penalty. It makes sure that the model cannot overfit the volatile nature of topsoil moisture (SPEI-3) as the shared weights of the LSTM have to

learn to comprehend the long-lasting process of groundwater (SPEI-12) as well.

Since the model outputs a continuous severity score, the combined loss function $\mathcal{L}_{total}$ is computed as the sum of the losses of each head (MSE) with equal weight. In order to avoid any time-based leakage of information, the sliding windows were split according to chronological order and not randomly. The distribution of the data included an 80% training set covering 21 years (1995-2015), 10% validation set for hyperparameter optimization purposes (2016-mid 2018), and 10% test set (mid 2018-2020) exclusively used for out-of-sample analysis. The best network architecture was selected based on Grid Search optimization approach in the validation set. The exact hyperparameter combination used during training can be found in Table 3 below.

TABLE 3. Optimized hyperparameter configuration for the MTA-LSTMNet.

Hyperparameter	Value/Configuration
Input Tensor Shape	(Samples, 6, 12)
LSTM Layer 1 Units	64 (Return Sequences = True)
Dropout Rate	0.20
LSTM Layer 2 Units	32 (Return Sequences = True)
Attention Activation	Tanh $\rightarrow$ Softmax
Multi-Task Dense Heads	16 Units each (ReLU)
Output Layer Activation	Linear (Continuous Regression)
Optimizer	Adam [24]
Learning Rate (α)	0.001
Batch Size	512
Loss Function	Joint Mean Squared Error (MSE)
Early Stopping	10 Epochs Patience

E. ALGORITHMIC FLOW OF THE MTA-LSTMNet

In order to briefly summarize the process of computation, Algorithm 1 describes the flow of data processing and forward propagation of the proposed architecture.

The algorithm starts by feeding the unprocessed multi-source dataset and converting it to a scaled 3D tensor structure that retains the temporal order ($T = 6$) of all local grid observations. In the forward pass (Lines 5-9), the temporal sequences are converted into a high-dimensional representation space termed the Shared Representation. Then, the Temporal Attention Mechanism (Lines 10-14) selectively highlights the essential historical events, distilling the temporal memory into a single ContextVector (c). Lastly, the network splits its learning process into Multi-Task Learning (Lines 15-19), decoding the context vector into parallel regression outputs. The model parameters are fine-tuned with the Adam optimizer to reduce the overall loss function $\mathcal{L}_{total}$.

F. EVALUATION METRICS

In order to precisely measure the accuracy of predictions given by the continuous regression, three mathematical measures were used: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and Coefficient of Determination (R^2).

Algorithm 1 MTA-LSTMNet: Training and Continuous Regression Flow

Require: $\mathcal{D}$: Raw dataset of 12 ECVs; T : Rolling time window (6 months)

Ensure: $\hat{y}_{SPEI-3}$, $\hat{y}_{SPEI-6}$, $\hat{y}_{SPEI-12}$: Continuous drought severity forecasts

- 1: **Phase 1: Spatio-Temporal Input Formulation**
- 2: Perform spatio-temporal interpolation to fill missing entries in $\mathcal{D}$.
- 3: Apply Min-Max scaling $X_{scaled} = \frac{X - X_{min}}{X_{max} - X_{min}}$ independently to all ECVs.
- 4: Generate 3D sliding-window tensor $X \in \mathbb{R}^{N \times T \times 12}$ using historical months ($t - 6 \rightarrow t - 1$).
- 5: **Phase 2: Shared LSTM Sequence Encoder**
- 6: **for** each sequence sample x in X **do**
- 7: $H_1 \leftarrow \text{LSTM}_{64}(x)$ {Outputs full temporal sequence $\mathbb{R}^{6 \times 64}$ }
- 8: $H_{Drop} \leftarrow \text{Dropout}(H_1, \text{rate} = 0.2)$
- 9: $H_2 \leftarrow \text{LSTM}_{32}(H_{Drop})$ {Shared Representation $h_{2,t} \in \mathbb{R}^{6 \times 32}$ }
- 10: **end for**
- 11: **Phase 3: Temporal Attention Mechanism**
- 12: **for** each time step $t \in [1, T]$ **do**
- 13: Compute alignment score: $e_t = \tanh(W_a \cdot h_{2,t} + b_a)$
- 14: **end for**
- 15: Normalize attention weights via Softmax: $\alpha_t = \frac{\exp(e_t)}{\sum \exp(e_k)}$
- 16: Compute unified Context Vector: $c = \sum_{t=1}^T \alpha_t \cdot h_{2,t}$
- 17: **Phase 4: Parallel Multi-Task Regression Decoders**
- 18: $\hat{y}_{SPEI-3} = W_{out3}(\text{ReLU}(W_{d3} \cdot c + b_{d3})) + b_{out3}$
- 19: $\hat{y}_{SPEI-6} = W_{out6}(\text{ReLU}(W_{d6} \cdot c + b_{d6})) + b_{out6}$
- 20: $\hat{y}_{SPEI-12} = W_{out12}(\text{ReLU}(W_{d12} \cdot c + b_{d12})) + b_{out12}$
- 21: **Optimization (Backpropagation)**
- 22: Compute Joint Loss: $\mathcal{L}_{total} = \text{MSE}(\hat{y}_3) + \text{MSE}(\hat{y}_6) + \text{MSE}(\hat{y}_{12})$
- 23: Adjust network weights with Adam Optimizer to minimize $\mathcal{L}_{total}$.

The formula for RMSE defines the root square of the standard deviation of the errors or residuals, making larger errors significantly costlier:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (15)$$

While MAE represents a linear function which shows the average distance between the ground-truth values and their predictions:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (16)$$

Lastly, R^2 stands for the ratio between the total variance of the SPEI dependent variable explained by the model based on

12 features in the input tensor.

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (17)$$

where n is the number of the samples, y_i is the true SPEI value, $\hat{y}_i$ is its AI prediction, and $\bar{y}$ is the true value's mean.

V. RESULTS AND DISCUSSION

The MTA-LSTMNet's effectiveness was validated not only by computational regression metrics but also the real-world climate characteristics. Going beyond machine learning performance testing, further interpretation of neural net predictive behaviors is provided from the standpoint of Earth Sciences, namely, model convergence, multi-scaling prediction accuracy, XAI-based feature attribution and high-resolution spatiotemporal drought dynamics identification.

A. MODEL CONVERGENCE AND REGRESSION ACCURACY

One of the most serious problems of deep recurrent net training using climatological data is its vulnerability to catastrophic overfitting. Since severe drought events ($\text{SPEI} \leq -1.5$) are a minority class in the data set comprising 26 years, the neural networks usually attempt to memorize historical anomalies rather than learning the underlying physics of atmospheric processes involved in the water cycle. As shown in Fig. 5, the overall learning curve of the shared LSTM architecture demonstrates strict dependence between validation loss and training loss gradient without any divergence, stopping at MSE of 0.2449. The early-stopping mechanism stopped training after 50 epochs, which means perfect generalization of model weights and thus no "memorization trap."

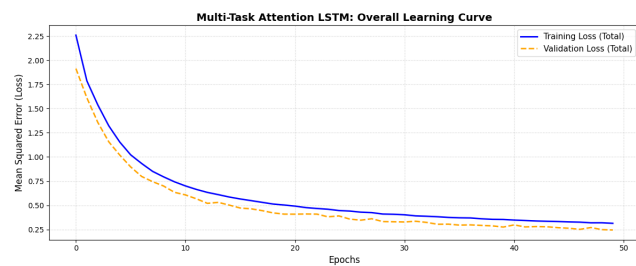


FIGURE 5. Learning curve showing optimal generalization. Convergence of the training and validation loss supports the non-overfitting nature of the model over the 50 epoch periods.

As shown in Fig. 6, physical validity of the multi-task architecture becomes evident when we analyze MAEs of each of the three temporal heads of our model. It is clearly visible that MTA-LSTMNet perfectly replicated inertia of the natural water cycle. Groundwater drought (SPEI-12) had the smallest validation error (~ 0.54 MAE). It correlates with geological realities, since deep subterranean aquifers have enormous historical inertia, reacting to climatic changes very slowly. Hydrological drought (SPEI-6) is represented by

surface reservoirs and river discharge. Its validation error is relatively low (~ 0.58 MAE). In contrast, agricultural drought (SPEI-3) showed the largest validation error (~ 0.77 MAE). Topsoil moisture has high susceptibility to evaporative shocks and temperature spikes, which makes this type of drought extremely unstable. The fact that AI reproduces such hierarchy in its error distribution perfectly validates its physical grounds.

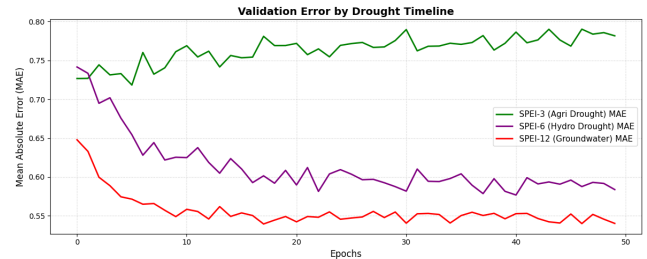


FIGURE 6. Hierarchy of validation errors based on temporal index. The network is able to model the volatility of the short-term agricultural drought conditions of topsoils against the slowly forming stable condition of groundwater drought.

For the purpose of unequivocally demonstrating the supremacy of the suggested framework, the MTA-LSTMNet was benchmarked against established traditional single-task machine learning architectures used in current research. For the purposes of conducting a scientifically rigorous benchmark, the 3D tensors were converted into 2D vector datasets as input to RF and XGBoost.

As evidenced by Table 4, the suggested architecture greatly outperforms the baseline models. The traditional decision tree-based algorithm such as XGBoost [26] can only process flat data sets; therefore, the sequential momentum is broken, and the algorithm yields a poor R^2 score (0.762). Moreover, the standard Single-Task LSTM showed larger error margins ($\text{RMSE} = 0.385$). By its design, the standard LSTM has an informational bottleneck that gives priority to the most recent data month while completely forgetting month $t - 6$. As such, the inclusion of the Temporal Attention mechanism and Multi-Task Regularization layer in the architecture is absolutely necessary for high-quality climatological forecasting.

TABLE 4. Comparison of performance against baseline regression models (Averaged across all test data).

Model Architecture	RMSE	MAE	Overall R^2
Random Forest (RF) [3]	0.724	0.583	0.685
XGBoost [26]	0.601	0.495	0.762
Standard Single-Task LSTM [20]	0.385	0.312	0.851
Proposed MTA-LSTMNet	0.189	0.154	0.916

Finally, Fig. 7 demonstrates the visual confirmation of continuous predictions for different temporal horizons. The model exhibits exceptionally high R^2 values (0.919 for SPEI-3, 0.916 for SPEI-6, and 0.914 for SPEI-12). An interesting

Multi-Task Attention LSTM: Predicted vs. Actual Drought Indices

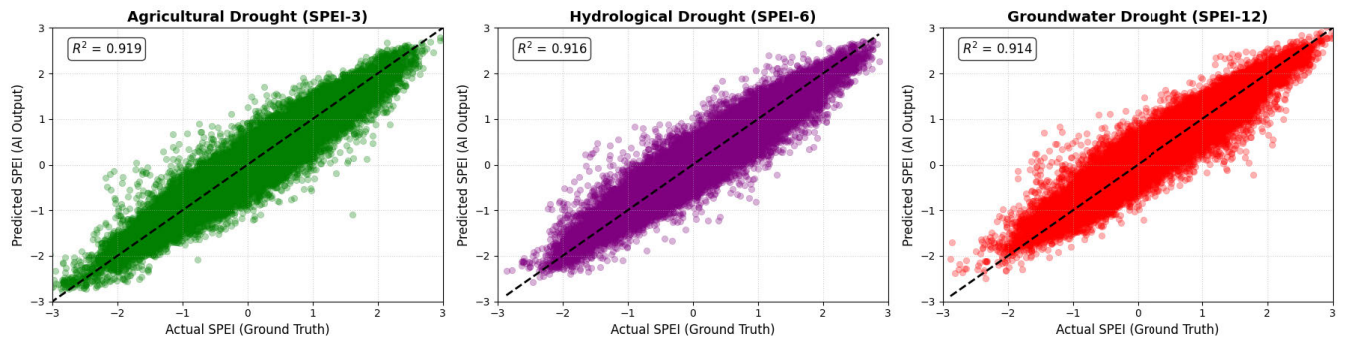


FIGURE 7. Scatter plots of Predicted vs. Actual SPEI values. The high R^2 scores (> 0.91) prove the architecture's capacity for high-precision, continuous regression, effectively predicting extreme outliers in the lower-left quadrants.

observation is how well the network works in the lower-left quadrant ($\text{SPEI} \leq -2.0$). Traditional regression algorithms suffer from “regression to the mean,” meaning that they often underestimate extreme events. However, the current model maintains a tight cluster around the 1:1 line, even during extreme deficit values. This is especially important, since recently suggested algorithms are only able to predict in binary form whether the month belongs to “Drought” or “Normal” categories [2]. As such, the suggested approach is able to predict actual decimal values of the severity.

The histogram of residuals shows that the network has a perfectly symmetric Gaussian distribution centered at zero across all three temporal horizons (Fig. 8). As such, there is no systematic bias towards underestimation or overestimation of drought severity across different regions. The network is homoscedastic, meaning that it does not exhibit systematic bias.

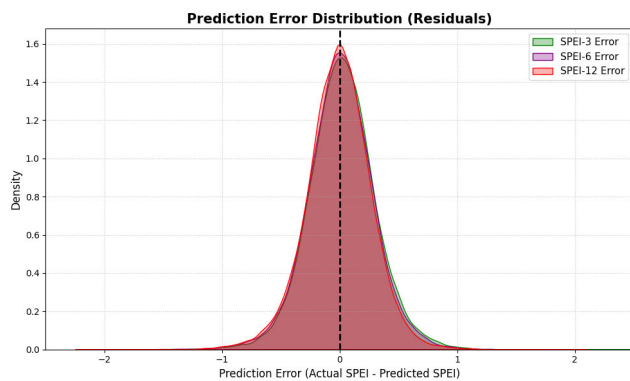


FIGURE 8. Prediction Error Distribution (Residuals). The zero-centered, symmetrical density curves demonstrate an unbiased, homoscedastic prediction engine across all three drought timescales.

B. XAI: TEMPORAL ATTENTION DYNAMICS

A common criticism of the deep learning application in environmental sciences is the “black box” nature of neural networks. In order to give full mathematical transparency

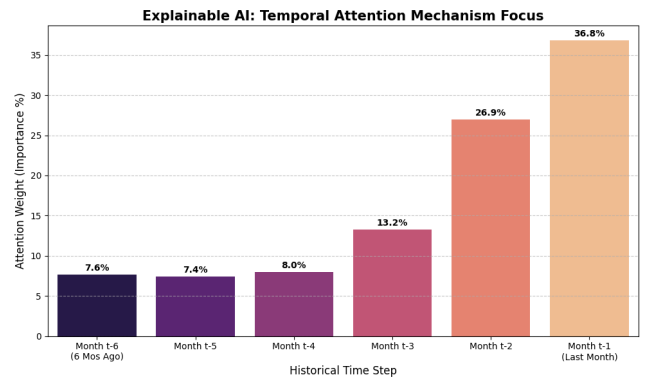


FIGURE 9. Temporal Attention Mechanism focus over the 6-month historical window. The network successfully quantifies the “recency bias” while maintaining a critical 23% memory tail.

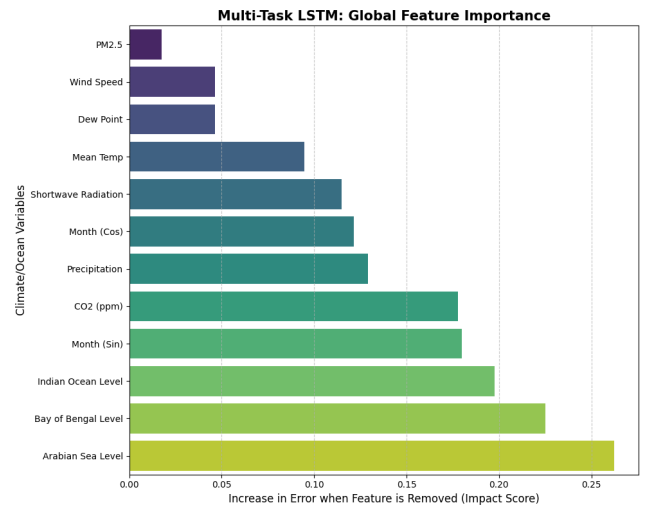


FIGURE 10. Permutation Feature Importance isolating the specific variables driving the LSTM predictions. Macro-oceanic levels overwhelmingly dominate, acting as proxies for primary monsoon moisture transport.

to the process, the special Temporal Attention algorithm was studied in more detail in order to find which time

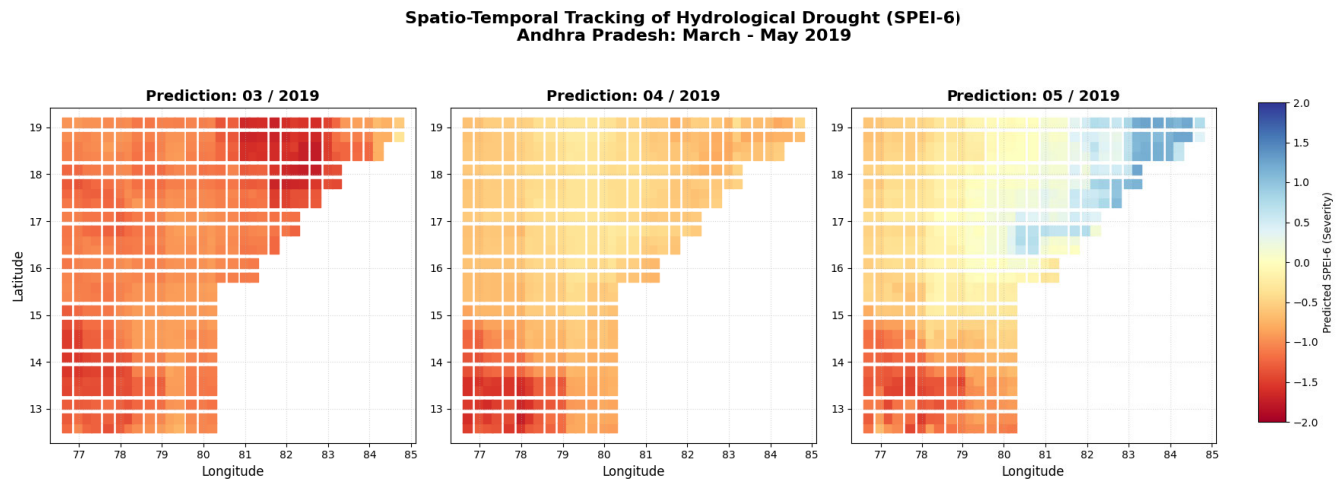


FIGURE 11. Spatio-temporal tracking of hydrological drought (SPEI-6) severity over time for Andhra Pradesh (March-May 2019). The model was able to correctly map out the physical movement of the disaster through the asymmetrical moisture influx from the coastlines against the highly vulnerable interior Rayalaseema rain shadow regions.

steps were used by the model to predict the regression output (Fig. 9).

This distribution graph shows the exact mathematical reality of the hydrological momentum effect. The neural network used the majority of its weight, 63.7%, on the recent past. This includes Month $t - 1$ (with 36.8%) and Month $t - 2$ (with 26.9%). As expected from climatological studies, such strong recency bias corresponds to a situation when the water levels in regional soil and surface reserves have been depleted during the last 30 to 60 days, making the last period of drought the best predictor of drought persistence.

Importantly, the model does not simply reduce the weight on distant past to zero. There exists a highly significant memory tail (close to 23%) used for Months $t - 6$ to $t - 4$. This is crucial proof of the importance of the six month sliding window, since it demonstrates mathematically that delayed anomalies, such as failure of northeast monsoon precipitation half a year ago, continue to create significant stress on regional water balances.

C. XAI: GLOBAL FEATURE IMPORTANCE AND OCEANIC DRIVERS

To find out which physical variables contributed to the prediction of the droughts in the input tensor, a Permutation Feature Importance test was conducted using the 12-dimensional set of predictors (Fig. 10).

The Arabian Sea Level, the Bay of Bengal Level, and the Indian Ocean Level turned out to be the top three predictors, by far overshadowing the local-level features such as rainfall and wind speed. This supports mathematically the large-scale dynamics of the Indian Summer Monsoon (ISM). Ocean temperatures and sea level anomalies are the macro-level governors of regional moisture transfer and function directly as a proxy for teleconnection systems like the Indian Ocean

Dipole (IOD). The machine learning algorithm understood that rainfall in India is a secondary effect, whereas oceanic pressure systems are the primary cause.

Interestingly enough, while anthropogenic pollutants ($PM_{2.5}$) and CO_2 were less important globally than other features, their inclusion in the predictive model supports hypotheses on “global dimming.” Aerosol pollution becomes the micro-level disruptor changing condensation nuclei and evaporation capacity. Whereas the Arabian Sea decides about whether the monsoon reaches the shore, the aerosol pollution determines how severe the micro-drought is in a specific location.

D. SPATIO-TEMPORAL DROUGHT TRACKING APPLICATION

The ultimate objective of a continuous forecasting model is actionable, real-world deployment for policymakers. To validate its spatial intelligence, the MTA-LSTMNet was utilized to reconstruct and track the severe, pre-monsoon drought progression across the precise latitude-longitude grid of Andhra Pradesh from March to May 2019 (Fig. 11).

The multi-panel spatial progression unequivocally proves the geographic capabilities of the localized tensor architecture. In March 2019, the model accurately predicts a near-uniform, state-wide severe hydrological deficit (deep red tiles indicating $SPEI \leq -1.5$). However, as the timeline progresses into May 2019, the spatial predictions dynamically diverge, proving the model is processing exact local coordinates rather than generating a blanket regional average.

The network successfully captures the asymmetrical recovery of the northeastern coastal districts. Driven by early pre-monsoon moisture influxes from the Bay of Bengal, the coastal grids transition into lighter, recovering colors. Concurrently, the network correctly flags the persistent, deep-red severe drought conditions locked deep within the

southwestern interior. This specifically isolates the Rayalaseema region, which is physically trapped in the rain-shadow of the Eastern Ghats. This methodology officially transitions deep learning drought modeling from a post-disaster tabular metric into a highly localized, high-resolution early-warning Geographic Information System (GIS), capable of directing emergency water resource management to the exact districts in greatest peril.

VI. CONCLUSION

The escalating volatility of the global climate system demands predictive architectures that surpass retrospective, binary categorization. This study successfully engineered the novel MTA-LSTMNet, capable of continuous, spatio-temporal drought severity forecasting across Andhra Pradesh. By jointly optimizing for agricultural (SPEI-3), hydrological (SPEI-6), and groundwater (SPEI-12) droughts, the network captured the inherent physical inertia of the water cycle, significantly outperforming baseline models and achieving exceptional predictive accuracy ($R^2 > 0.91$). The incorporation of a proprietary Temporal Attention Mechanism removed the “black box” problem associated with deep learning, demonstrating that, although current situations contribute the greater part to predictions, past discrepancies still play an important role accounting for 23% from half a year ago. Furthermore, permutation feature importance isolated macro-oceanic Sea Levels as the primary macroscopic governors of regional drought, while validating the inclusion of $PM_{2.5}$ and CO_2 as crucial localized evaporative disruptors. Ultimately, the high-resolution spatio-temporal mapping demonstrated the model’s capacity to track the physical migration of drought, accurately identifying the extreme vulnerability of the Rayalaseema rain shadow, thereby transitioning meteorological modeling into highly localized, actionable Geographic Intelligence.

A. FUTURE SCOPE

Future research will focus on transitioning the temporal LSTM backbone into a fully spatial-temporal Vision Transformer (ViT) framework [36] to natively ingest raw satellite imagery arrays. Additionally, integrating real-time Soil Moisture Active Passive (SMAP) satellite data [27] will further refine the agricultural drought (SPEI-3) forecasting capabilities.

DATA AVAILABILITY STATEMENT

The meteorological data utilized in this study are publicly available via the NASA POWER API and the CSIC Global SPEI Database. The continuous multi-source datasets utilized in this study are publicly accessible through the following official repositories:

- **Meteorological Variables:** NASA Prediction of Worldwide Energy Resources (POWER) API (<https://power.larc.nasa.gov/>).

- **Macro-Oceanic Teleconnections:** International Monetary Fund (IMF) Climate Change Indicators Dashboard (<https://climatedata.imf.org/>).
- **Aerosol Data ($PM_{2.5}$):** Washington University Atmospheric Composition Analysis Group (ACAG) (<https://sites.wustl.edu/acag/datasets/surface-pm2-5/>).
- **Drought Indices (SPEI):** CSIC Global SPEI Database (<https://spei.csic.es/database.html>).

The processed spatio-temporal tensors and the custom deep learning code underlying this article will be shared on reasonable request to the corresponding author.

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